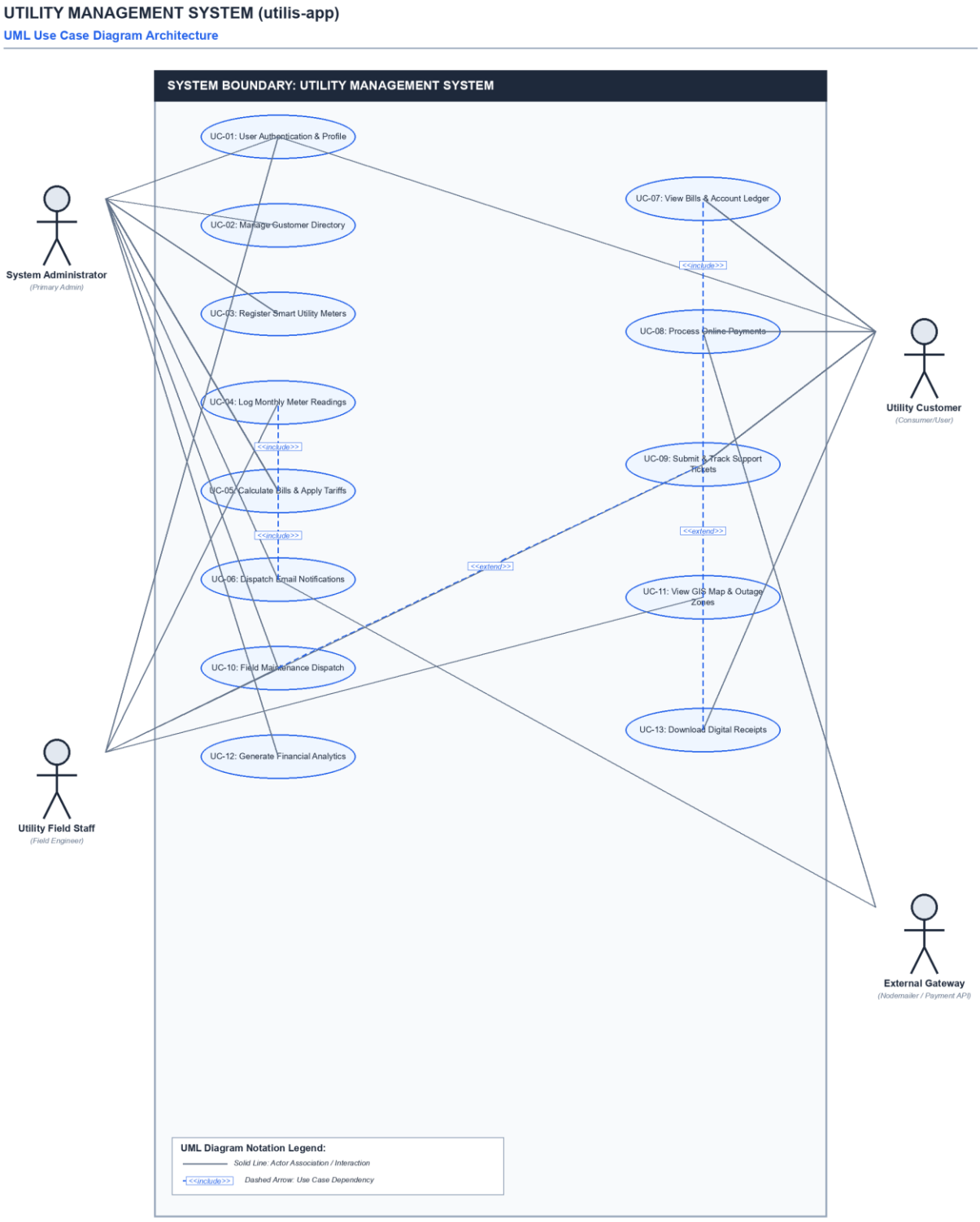
UTILITY MANAGEMENT SYSTEM (utilis-app)

UML Use Case Diagram Architecture | Student: Jasika Noor (ID: 22CSE002)



Use Case Diagram Elements & Actor Summary Legend

UC ID	Use Case Name	Associated Actor(s)	Core Functionality
UC-01	User Authentication & Profile	Admin, Field Staff, Customer	Secure portal login, self-registration, session handling, and profile updates.
UC-02	Manage Customer Directory	System Administrator	Create, edit, search customer profiles and link service addresses.
UC-03	Register Smart Utility Meters	System Administrator	Register utility meters (Electricity, Water, Gas) and assign to customers.
UC-04	Log Monthly Meter Readings	Field Staff, Customer	Log monthly consumption units and calculate delta (currReading - prevReading).
UC-05	Calculate Bills & Apply Tariffs	System Administrator	Automated billing engine computing bill amount based on tiered tariffs.
UC-06	Dispatch Email Notifications	Admin, External Nodemailer API	Automated Nodemailer email dispatch for new invoices and urgent payment reminders.
UC-07	View Bills & Account Ledger	Utility Customer	Consumer access to unpaid bills, historical ledgers, and payment due dates.
UC-08	Process Online Payments	Customer, External Payment API	Digital bill payment (bKash, Nagad, Bank) and automated bill status update.
UC-09	Submit & Track Support Tickets	Customer, Field Staff	Customer complaint filing with interactive comment threads and status updates.
UC-10	Field Maintenance Dispatch	Admin, Field Staff	Schedule emergency repair work orders, assign field technicians, and log resolutions.

UC ID	Use Case Name	Associated Actor(s)	Core Functionality
UC-11	View GIS Map & Outage Zones	Field Staff, Admin	Interactive Leaflet spatial mapping of customer meters and outage fault zones.
UC-12	Generate Financial Analytics	System Administrator	Executive revenue collection charts, utility breakdown, and financial reports.
UC-13	Download Digital Receipts	Utility Customer	Generate printable payment receipts featuring unique transaction reference IDs.